Factor-Ray Dephasing of a Prime-Target Residual: A Hyperbolic Mellin Large Sieve, Averaged-Shift Energy, and the Intra-Ray Boundary

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Abstract

Put $u(n) = \Lambda(n) - 1$, fix a real-valued $W \in C_c^1((0,\infty))$, and consider the large-source-factor transform

$$\mathcal{T}_{h,D}(X;t) = \sum_{\substack{n>D \\ m\geq 1}} u(n)\Lambda(mn+h)W(mn/X)(n/m)^{it}.$$

The target-free form $\mathcal{M}_D$, obtained by replacing $\Lambda(mn+h)$ by 1, is the first-moment model under a long smooth average of the shift. We study the centered transform $\mathcal{R}_{h,D} = \mathcal{T}_{h,D} - \mathcal{M}_D$.

Writing uniquely $n = ak$, $m = bk$, $(a,b) = 1$, compresses every equal-ratio class into a primitive factor ray. The resulting frequencies $\log(a/b)$ have spacing at least

$$\delta_X = 2\operatorname{arsinh}\left(\frac{1}{2\beta X}\right)$$

when $\operatorname{supp} W \subset [\alpha,\beta]$. The generalized Hilbert inequality of Montgomery and Vaughan therefore gives, for every fixed shift and every $T \geq 1$,

$$\int_{-T}^{T} |\mathcal{R}_{h,D}(X;t)|^2 dt = 2T\mathcal{E}_{h,D}(X) + O_W(X\mathcal{E}_{h,D}(X)),$$

where $\mathcal{E}_{h,D}$ is the exact squared ray energy. A raywise Cauchy ledger proves $\mathcal{E}_{h,D} \ll X(\log X)^4$. Consequently every prescribed fixed integer shift has mean-square $o(X^2)$ per Mellin frequency on bands whose length grows faster than $(\log X)^4$. This is frequency-density cancellation, not a fixed-frequency asymptotic. With a normalized Fejér kernel, distinct rays are removed exactly once $T\delta_X \geq 1$.

The ray energy itself becomes arithmetically evaluable after squaring first and then averaging the shift. For nonnegative $G \in C_c^1((0,\infty))$, $X \leq H \leq X^C$, and $1 \leq D \leq X^{1-\varepsilon}$, we prove

$$\begin{aligned}\sum_h G(h/H)\mathcal{E}_{h,D}(X) = {}& \frac{1}{2}HXI_0(G)I_2(W)\log H \\ & \times \big((\log X)^2 - (\log D)^2\big) \\ & + O\big(HX(\log X)^2\log\log(3X)\big)\,.\end{aligned}$$

The diagonal Λ^2 term supplies the extra logarithm. Every same-ray off-diagonal target correlation is handled only by a two-linear- forms Selberg upper sieve and an explicit

$O(X\log^2 X)$ source-ray ledger. This yields a genuine hard-band asymptotic for $T/X\to\infty$ and an exact Fejér-band asymptotic already at $T\asymp X$.

Finally, a determinant-layer formula identifies the unresolved mesoscopic band, and a fixed smooth endpoint selector shows that an $o(X^2)$ fixed-shift band error, together with a compatible filtered model and $o(X)$ Fourier tails, would recover the $m=1$ Hardy–Littlewood functional. Equal-ratio compression never separates the terms $\Lambda(abk^2+h)$ on one ray. Thus no fixed-shift prime-pair asymptotic, twin-prime lower bound, new level of distribution, or breach of the parity barrier is claimed.

Keywords: prime-target residual; Mellin mean square; Montgomery–Vaughan inequality; factor rays; averaged shifts; Selberg upper-bound sieve; sieve parity.

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1 Introduction and claim ledger

The factor-scale coordinate introduced in TPC–9 and TPC–10 records a factorization $r=mn$ by

$$u_{m,n}=\log(n/m).$$

At frequency zero, summing over all divisors gives the exact collapse

$$\sum_{n|r}(\Lambda(n)-1)=\log r-\tau(r),$$

which leads to a fixed-shift Titchmarsh evaluation [1, 10]. TPC–10 proved coarse two-sided factor localization and arbitrary fixed smooth localization on the Type I side. It also showed that finitely many Mellin coordinates do not stably determine a local factor-scale window, and proposed a continuum-band mean square as the next analytic target [8].

The present paper treats that target in two different quantifier regimes. Let $X\ge 2$, $h\in\mathbb{Z}$, and $D\ge 1$. Fix throughout

$$u(n):=\Lambda(n)-1,\qquad W\in C_c^1((0,\infty);\mathbb{R}),\qquad \operatorname{supp}W\subset[\alpha,\beta] \tag{1}$$

with $0<\alpha<\beta<\infty$. We extend $\Lambda(q)$ by zero for $q\le 0$. For any integrable function F on $(0,\infty)$, put

$$I_0(F):=\int_0^\infty F(v)\,dv,\qquad I_2(F):=\int_0^\infty |F(v)|^2\,dv. \tag{2}$$

All asymptotic statements are taken as $X\to\infty$; parameters declared fixed in a theorem are held fixed, while D,H, and T may vary in the displayed ranges. Define

$$\mathcal{T}_{h,D}(X;t):=\sum_{\substack{n>D\\ m\ge 1}}u(n)\Lambda(mn+h)W(mn/X)(n/m)^{it}, \tag{3}$$

$$\mathcal{M}_D(X;t):=\sum_{\substack{n>D\\ m\ge 1}}u(n)W(mn/X)(n/m)^{it}, \tag{4}$$

$$\mathcal{R}_{h,D}(X;t):=\mathcal{T}_{h,D}(X;t)-\mathcal{M}_D(X;t). \tag{5}$$

Thus $\mathcal{R}_{h,D}$ is obtained by replacing the target factor $\Lambda(mn+h)$ by $\Lambda(mn+h)-1$. The subtraction is algebraic for every fixed h. It becomes a genuine first-moment model only after the long shift average proved in proposition 4.1.

1.1 What is proved

There are four main outputs.

(i) Every factor pair admits a unique representation

$$n = ak, \qquad m = bk, \qquad (a, b) = 1.$$

Equal ratios are thereby compressed into one primitive ray (a, b). Distinct ray frequencies $\log(a/b)$ have spacing $\gg_W X^{-1}$. Applying the continuous mean-value theorem of Montgomery and Vaughan [7] gives an exact fixed-shift diagonal law up to an $O(X)$ resolution error.

(ii) The ray energy has the unconditional fixed-shift bound

$$\mathcal{E}_{h,D}(X) \ll_{C,W} X(\log X)^4 \qquad (|h| \le X^C).$$

Hence a band of length $T \gg (\log X)^4$ gives $o(X)$ cancellation in root-mean-square per frequency, for every prescribed shift. This does not evaluate any specified Mellin frequency.

(iii) After the error is squared, a smooth average over shifts $h \asymp H$, $H \ge X$, permits a full arithmetic evaluation of the ray energy. The $k = \ell$ target diagonal has size $H \log H$. Same-ray terms $k \ne \ell$ require prime-pair upper bounds, but not prime-pair asymptotics: a Selberg upper sieve costs only $H \log\log X$, and a source-ray ledger sums their coefficients to $O_W(X \log^2 X)$. The diagonal therefore wins by one logarithm.

(iv) A compact-Fourier-support kernel resolves the factor rays exactly at the natural threshold $T \asymp X$. Below that threshold, the surviving interactions are indexed by the integer determinant $ad - bc$. At the opposite endpoint, a fixed smooth scale window can select $m = 1$ exactly on every product band of multiplicative width less than four. This gives a quantitative $o(X^2)$ strength threshold for any fixed-shift continuum-band model that could imply the corresponding Hardy–Littlewood functional.

The generalized Hilbert inequality is the only input in the fixed-shift spectral theorem. The averaged-shift energy theorem additionally uses the prime number theorem and a standard two-linear-forms Selberg upper sieve. It does not import an averaged Hardy–Littlewood asymptotic. For comparison, Theorem 1.3(i) of Matomäki et al. [5] evaluates two-point prime correlations for almost all shifts in the substantially shorter range $X^{8/33+\varepsilon} \le H \le X^{1-\varepsilon}$; TPC–7 already transferred that result to a different complete defect [9]. The later higher-uniformity theorem of Matomäki et al. [6, Theorem 1.5] treats higher orders while retaining the earlier two-point input. Our theorem instead evaluates a continuum of Mellin frequencies after a longer second-moment shift average.

1.2 What is not proved

The order of averaging is essential. We first form $|\mathcal{R}_{h,D}(X; t)|^2$ and only then average h. A coherent average of $\mathcal{R}_{h,D}$ before squaring is much easier and says nothing about a typical shift. Likewise, an average over Mellin frequencies does not select $t = 0$ or any other prescribed frequency.

On one primitive ray, the target arguments are

$$abk^2 + h.$$

No amount of separation between distinct values of a/b decorrelates different k on the same ray. In particular, the ray $(1, 1)$ already contains values of $k^2 + h$, while the factor endpoint

$b = 1$, $k = 1$ contains the fixed-shift prime-pair functional. We neither estimate these objects at a prescribed h nor assert that such an estimate is impossible. No twin-prime lower bound, fixed-shift Hardy–Littlewood asymptotic, new Type II estimate, new distribution level, or parity-barrier breakthrough follows.

For later reference, the logical ledger is displayed in table 1.

Input	Proved output	Not obtained
Reduced factor ratios + Montgomery–Vaughan	Fixed-h continuum mean square and exact ray resolution	Arithmetic evaluation of one fixed-h ray energy
Long smooth shift average + PNT + upper sieve	Explicit averaged ray-energy and band asymptotics	Any prescribed individual shift
Compact-Fourier kernel	Exact removal of distinct rays at $T \asymp X$	Separation of different k on one ray
Endpoint scale selector	Exact $m = 1$ reduction and an L^2 strength threshold	The Hardy–Littlewood value of that endpoint

Table 1: Every implication used in the paper is one-way.

2 Primitive factor rays and logarithmic separation

The Mellin phase depends only on the reduced ratio of the two factors. This elementary observation is the structural starting point of the paper.

Proposition 2.1 (Exact factor-ray compression)**.** *For every $m, n \geq 1$, write*

$$k = (m, n), \qquad a = n/k, \qquad b = m/k.$$

Then $(a, b) = 1$, the representation $n = ak$, $m = bk$ is unique, and

$$\mathcal{R}_{h,D}(X;t) = \sum_{\substack{a,b\geq 1 \\ (a,b)=1}} c_{a,b}(h;X,D)(a/b)^{it}, \tag{6}$$

where

$$c_{a,b}(h;X,D) := \sum_{\substack{k\geq 1 \\ ak>D}} u(ak)(\Lambda(abk^2+h)-1)W(abk^2/X). \tag{7}$$

Only rays with $ab \leq \beta X$ occur. The raw transform $\mathcal{T}_{h,D}$ has the same decomposition with the target factor in (7) replaced by $\Lambda(abk^2+h)$.

Proof. The greatest-common-divisor decomposition is unique, and

$$mn = abk^2, \qquad \frac{n}{m} = \frac{a}{b}.$$

Substitution in (5) gives (6). If a term is nonzero, then $abk^2/X \in \operatorname{supp} W \subset [\alpha, \beta]$, hence $ab \leq \beta X$. □

Definition 2.2 (Ray energy)**.** The centered factor-ray energy is

$$\mathcal{E}_{h,D}(X) := \sum_{\substack{a,b\geq 1 \\ (a,b)=1}} |c_{a,b}(h;X,D)|^2. \tag{8}$$

The analogous raw energy, with Λ in place of $\Lambda - 1$ at the target, is denoted by $\mathcal{E}^{\mathrm{raw}}_{h,D}(X)$.

Equal ratios have now been combined before any mean-square theorem is applied. This is indispensable: separate pairs on one ray have exactly the same Mellin frequency and cannot be made orthogonal by increasing the length of the frequency interval.

Lemma 2.3 (Hyperbolic log-ratio spacing)**.** *Let* $(a,b) = (c,d) = 1$*, let* $ab, cd \le \beta X$*, and suppose* $(a,b) \ne (c,d)$*. Then*

$$\left|\log \frac{a}{b} - \log \frac{c}{d}\right| \ge \delta_X, \qquad \delta_X := 2\,\mathrm{arsinh}\left(\frac{1}{2\beta X}\right). \tag{9}$$

In particular, $\delta_X \asymp_W X^{-1}$.

Proof. Put $A = ad$ and $B = bc$. Since the two fractions are distinct, $|A - B| \ge 1$. Moreover,

$$AB = abcd = (ab)(cd) \le (\beta X)^2.$$

The exact identity

$$2\sinh\left(\frac{1}{2}\left|\log \frac{A}{B}\right|\right) = \frac{|A-B|}{\sqrt{AB}}$$

therefore gives

$$2\sinh\left(\frac{1}{2}\left|\log \frac{a/b}{c/d}\right|\right) \ge \frac{1}{\beta X}.$$

Applying the increasing inverse hyperbolic sine proves (9). □

Remark 2.4 (Why the scale is X^{-1} rather than X^{-2})*.* The cross-products ad and bc could individually be large, but their product is at most $(\beta X)^2$. Near equality they are therefore both $O_W(X)$, and the integer determinant $ad - bc$ supplies a gap of order X^{-1}. Ignoring the hyperbolic constraints $ab, cd \ll X$ would lose this improvement.

Remark 2.5 (Near-sharp pairs)*.* The rays $(N, 1)$ and $(N - 1, 1)$ have determinant one and log gap

$$\log \frac{N}{N-1} = \frac{1}{N} + O(N^{-2}).$$

Taking $N \asymp X$ shows that no uniform separation scale larger than a constant multiple of X^{-1} is available for arbitrary hyperbolic coefficients. Thus the X^{-1} lower-bound scale is sharp up to constants; equivalently, no uniform resolution time $o(X)$ is possible. This threshold is geometric, before any prime weight is inserted.

3 Fixed-shift continuum dephasing

We first keep the shift completely prescribed. The only cancellation in this section comes from integration in the Mellin frequency.

Theorem 3.1 (Fixed-shift factor-ray mean square)**.** *Let* $I \subset \mathbb{R}$ *be any bounded interval. Then*

$$\int_I |\mathcal{R}_{h,D}(X;t)|^2 dt = |I|\mathcal{E}_{h,D}(X) + O_W(X\mathcal{E}_{h,D}(X)). \tag{10}$$

In particular,

$$\int_{-T}^{T} |\mathcal{R}_{h,D}(X;t)|^2 dt = 2T\mathcal{E}_{h,D}(X) + O_W(X\mathcal{E}_{h,D}(X)). \tag{11}$$

The same statements hold for $\mathcal{T}_{h,D}$ *and* $\mathcal{E}^{\mathrm{raw}}_{h,D}$*.*

Proof. By proposition 2.1, the centered transform is an exponential polynomial with coefficients $c_{a,b}$ and distinct real frequencies $\lambda_{a,b} = \log(a/b)$. The frequencies are δ_X-separated by lemma 2.3. Corollary 2, equations (1.8)–(1.9), of the generalized Hilbert inequality of Montgomery and Vaughan [7] gives

$$\int_I \left|\sum_\rho c_\rho e^{it\lambda_\rho}\right|^2 dt = |I|\sum_\rho |c_\rho|^2 + O\left(\delta_X^{-1}\sum_\rho |c_\rho|^2\right).$$

Translation of I only modulates the coefficients and does not change their squared norm. Since $\delta_X^{-1} \ll_W X$, this is (10). The raw form has the same frequency set. $\square$

The arithmetic coefficients have a substantially better energy bound than one obtains by squaring the number of factor pairs.

Lemma 3.2 (A source-ray square ledger)**.** *Uniformly for $Y \geq 2$,*

$$\sum_{ak^2 \leq Y} \frac{(\Lambda(ak)-1)^2}{ak} \ll (\log(2Y))^2. \tag{12}$$

Proof. Use $(\Lambda - 1)^2 \leq 2(1+\Lambda^2)$. The contribution of 1 is

$$\sum_{k\leq\sqrt{Y}} \frac{1}{k} \sum_{a\leq Y/k^2} \frac{1}{a} \ll (\log(2Y))^2.$$

If $\Lambda(ak) \neq 0$, write $ak = p^j$. Every representation has $k = p^\ell$, $a = p^{j-\ell}$, and the extra constraint is $p^{j+\ell} \leq Y$. Dropping that constraint except for $p^j \leq Y$ gives

$$\sum_{p^j\leq Y} (j+1)\frac{(\log p)^2}{p^j} \ll (\log(2Y))^2.$$

The $j = 1$ part has this order by partial summation and the Chebyshev bounds; the sum over $j \geq 2$ is bounded. $\square$

Proposition 3.3 (Uniform fixed-shift energy bound)**.** *Fix $C \geq 1$. Uniformly for $|h| \leq X^C$ and every $D \geq 1$,*

$$\mathcal{E}_{h,D}(X) + \mathcal{E}^{\mathrm{raw}}_{h,D}(X) \ll_{C,W} X(\log(2X))^4. \tag{13}$$

Consequently,

$$\int_I |\mathcal{R}_{h,D}(X;t)|^2 dt \ll_{C,W} (|I| + X)X(\log(2X))^4. \tag{14}$$

Proof. For a ray (a, b) let

$$K_{a,b} := \{k \geq 1 : ak > D,\ abk^2/X \in \operatorname{supp} W\}.$$

The support interval has fixed multiplicative width. Hence, for every $k \in K_{a,b}$,

$$|K_{a,b}| \ll_W k. \tag{15}$$

Cauchy's inequality and $|\Lambda(q) - 1| \ll_C \log(2X)$ for the target arguments in question give

$$|c_{a,b}|^2 \ll_{C,W} (\log(2X))^2 \sum_{k\in K_{a,b}} k\, u(ak)^2 |W(abk^2/X)|^2.$$

For fixed a, k, the number of b in the support is

$$\ll_W \frac{X}{ak^2}. \tag{16}$$

The possible additive 1 is absorbed because nonemptiness implies $ak^2 \ll_W X$. Summing first over b therefore gives

$$\mathcal{E}_{h,D}(X) \ll_{C,W} X(\log(2X))^2 \sum_{ak^2 \ll_W X} \frac{u(ak)^2}{ak}.$$

Apply lemma 3.2. The raw target obeys the same pointwise bound. Finally insert (13) into theorem 3.1. □

Corollary 3.4 (Density-one Mellin-frequency cancellation)**.** *Let I_X be any intervals with lengths L_X satisfying*

$$\frac{L_X}{(\log X)^4} \longrightarrow \infty.$$

For every fixed h and every $D = D(X)$,

$$\frac{1}{L_X X^2} \int_{I_X} |\mathcal{R}_{h,D}(X;t)|^2 dt = o(1). \tag{17}$$

In particular, outside a subset of relative measure $o(1)$ in I_X, one has $\mathcal{R}_{h,D}(X;t) = o(X)$.

Proof. Divide (14) by $L_X X^2$:

$$\frac{1}{L_X X^2} \int_{I_X} |\mathcal{R}_{h,D}|^2 dt \ll_{h,W} \frac{(\log X)^4}{X} + \frac{(\log X)^4}{L_X}.$$

Both terms tend to zero. The density statement follows from Chebyshev's inequality, choosing an $o(X)$ threshold slowly enough. □

Remark 3.5 (What the corollary does not select)*.* The exceptional set in corollary 3.4 may contain every prescribed frequency, including $t = 0$, at every scale. Moreover, (17) is an average per unit frequency; it is not the absolute $o(X^2)$ band error needed by the endpoint recovery criterion in section 7.

There is also an unconditional lower floor. It will be useful when interpreting the averaged theorem.

Proposition 3.6 (Nonzero ray-energy floor)**.** *Suppose $\beta < 4\alpha$, let $D < \alpha X$, and fix an integer h. If $W \not\equiv 0$, then*

$$\mathcal{E}_{h,D}(X) \gg_{h,W} X \log X. \tag{18}$$

The same lower bound holds for the raw ray energy.

Proof. On a ray $(a, 1)$ with $a/X \in \operatorname{supp} W$, the support condition $\beta < 4\alpha$ excludes every $k \geq 2$. Hence

$$c_{a,1} = u(a)(\Lambda(a + h) - 1)W(a/X).$$

There is an absolute $c_* > 0$ such that $u(a)^2 \geq c_*$ for every $a \geq 1$: off prime powers the value is 1, and on prime powers it is $(\log p - 1)^2$, whose minimum over primes is positive. The prime number theorem and partial summation give

$$\sum_a (\Lambda(a + h) - 1)^2 |W(a/X)|^2 = X \log X \int_0^\infty |W(v)|^2 dv + O_{h,W}(X).$$

Keeping only these distinct rays proves (18). Replacing $\Lambda - 1$ by Λ at the target uses the analogous second-moment formula for Λ^2. □

4 Arithmetic evaluation after averaging the shift

We now square first and average the shift afterwards. Let

$$G \in C_c^1((0,\infty)), \qquad G \ge 0,$$

with I_0 and I_2 as in (2). The shift sum below is over all integers; the support of $G(h/H)$ makes it finite and positive for large H.

4.1 The first-moment model

Proposition 4.1 (Long-shift first moment)**.** *Fix $C \ge 1$. Uniformly for $X \le H \le X^C$, every real t, every $D \ge 1$, and every $A > 0$,*

$$\frac{1}{H}\sum_h G(h/H)\mathcal{T}_{h,D}(X;t) = I_0(G)\mathcal{M}_D(X;t) + O_{A,C,W,G}\Big(X(\log X)^{-A}\Big). \tag{19}$$

Equivalently, the same normalized average of $\mathcal{R}_{h,D}$ is $O_A(X(\log X)^{-A})$.

Proof. For $r \in [\alpha X, \beta X]$, the smooth prime number theorem gives, uniformly in r and H in the stated range,

$$\sum_h G(h/H)(\Lambda(r+h)-1) \ll_{A,C,W,G} H(\log X)^{-A}. \tag{20}$$

Indeed, after the change of variable $q = r + h$, the weight $G((q-r)/H)$ has scale H and is supported on $q \asymp_{C,G} H$. The classical zero-free-region prime number theorem is more than sufficient.

The remaining source absolute mass satisfies

$$\sum_{\substack{n>D\\ m\ge 1}} |u(n)W(mn/X)| \ll_W X \sum_{n \ll_W X} \frac{1+\Lambda(n)}{n} + O_W(X) \ll_W X\log(2X).$$

Insert (20), request two additional powers of $\log X$, and divide by H. The Mellin phase has modulus one, so the estimate is uniform in t. □

The coherent average in proposition 4.1 is not a theorem about most shifts. That requires the second moment developed next.

4.2 The diagonal inputs

Lemma 4.2 (Translated target second moment)**.** *Uniformly for $r \in [\alpha X, \beta X]$ and $X \le H \le X^C$,*

$$\sum_h G(h/H)(\Lambda(r+h)-1)^2 = HI_0(G)\log H + O_{C,W,G}(H). \tag{21}$$

The same formula, with the same leading term, holds with $(\Lambda(r+h)-1)^2$ replaced by $\Lambda(r+h)^2$.

Proof. Expand $(\Lambda-1)^2 = \Lambda^2 - 2\Lambda + 1$. Smooth partial summation and the prime number theorem give

$$\sum_h G(h/H)\Lambda(r+h) = HI_0(G) + O_A(H(\log H)^{-A})$$

and

$$\sum_h G(h/H) = HI_0(G) + O_G(1).$$

For the square, first restrict to prime target values. One von Mangoldt factor supplies the prime number theorem, while the other is $\log H + O_{C,W,G}(1)$ on the support. Thus the prime contribution is $HI_0(G)\log H + O_{C,W,G}(H)$. Proper prime powers contribute $O_{C,W,G}(H^{1/2}(\log H)^2)$, which is absorbed. Combining the three pieces proves the centered formula. □

Lemma 4.3 (Source diagonal). *Uniformly for $1 \le D \le X^{1-\varepsilon}$,*

$$\begin{aligned} \mathcal{S}_D(X) &:= \sum_{\substack{n>D\\ m\ge 1}} u(n)^2 |W(mn/X)|^2 \\ &= \frac{1}{2} X I_2(W)\big((\log X)^2 - (\log D)^2\big) + O_W(X\log(2X)). \end{aligned} \tag{22}$$

Proof. Euler summation gives, uniformly in n,

$$\sum_{m\ge 1} |W(mn/X)|^2 = \frac{X}{n} I_2(W) + O_W(1). \tag{23}$$

The elementary weighted second moment

$$\sum_{n\le y} \frac{(\Lambda(n)-1)^2}{n} = \frac{1}{2}(\log y)^2 - \log y + O(1) \tag{24}$$

follows by expanding the square, using the classical zero-free-region prime number theorem for $\sum \Lambda(n)^2/n$ and $\sum \Lambda(n)/n$, and summing proper prime powers absolutely. Also $\sum_{n\le y}(\Lambda(n)-1)^2 \ll y\log(2y)$. Insert (23), subtract the ranges up to D, and absorb the fixed support endpoints into $O_W(X\log X)$. □

4.3 The same-ray off-diagonal ledger

Lemma 4.4 (Two-target upper sieve). *Let $r \ne r'$ lie in $[\alpha X, \beta X]$. Uniformly for $X \le H \le X^C$,*

$$\begin{aligned} &\sum_h G(h/H)\, |(\Lambda(r+h)-1)(\Lambda(r'+h)-1)| \\ &\qquad \ll_{C,W,G} H \prod_{\substack{p\mid r-r'\\ p>2}} \frac{p-1}{p-2} \ll_{C,W,G} H \log\log(3X). \end{aligned} \tag{25}$$

The same bound holds if either centered target factor is replaced by Λ.

Proof. The constant and one-prime terms cost $O_G(H)$. If both target values are prime, the Selberg upper-bound sieve for the two linear forms $h+r$ and $h+r'$ gives

$$\#\{h \asymp_G H : h+r,\ h+r' \text{ prime}\} \ll_G \frac{H}{(\log H)^2} \prod_{\substack{p\mid r-r'\\ p>2}} \frac{p-1}{p-2}.$$

Restoring the two logarithmic weights proves the first bound for the prime-prime part. If either target is a proper prime power, the aggregate contribution is $O_G(H^{1/2}(\log H)^2)$.

Finally,

$$\prod_{\substack{p\mid q\\ p>2}} \frac{p-1}{p-2} \ll \log\log(3|q|)$$

by the standard maximal-order comparison with the product over the smallest primes; here $0 < |r-r'| \ll_W X$. This proves (25). See, for example, Halberstam and Richert [2, Chapter 6] for the two-linear-forms upper sieve. □

Lemma 4.5 (Same-ray source off-diagonal)**.** *For every $D \ge 1$,*

$$\begin{aligned}\mathcal{L}_D(X) := \sum_{(a,b)=1} \sum_{\substack{k\neq\ell \\ ak,a\ell>D}} &|u(ak)u(a\ell)| \\ &\times |W(abk^2/X)W(ab\ell^2/X)| \ll_W X(\log(2X))^2.\end{aligned} \tag{26}$$

Proof. For a fixed ray use the set $K_{a,b}$ from proposition 3.3 and put $x_k = |u(ak)W(abk^2/X)|$. Then

$$\sum_{k\neq\ell} x_k x_\ell \le |K_{a,b}| \sum_k x_k^2 \ll_W \sum_{k\in K_{a,b}} kx_k^2$$

by (15). Summing over b and applying (16) yields

$$\mathcal{L}_D(X) \ll_W X \sum_{ak^2 \ll_W X} \frac{u(ak)^2}{ak}.$$

Now use lemma 3.2. □

4.4 The averaged ray-energy theorem

Theorem 4.6 (Averaged-shift ray energy)**.** *Fix $\varepsilon > 0$ and $C \ge 1$. Uniformly for*

$$1 \le D \le X^{1-\varepsilon}, \qquad X \le H \le X^C,$$

one has

$$\begin{aligned}\sum_h G(h/H)\mathcal{E}_{h,D}(X) = \frac{1}{2}&HXI_0(G)I_2(W)\log H \\ &\times \left((\log X)^2 - (\log D)^2\right) \\ &+ O_{\varepsilon,C,W,G}\Big(HX(\log(2X))^2 \log\log(3X)\Big).\end{aligned} \tag{27}$$

The same asymptotic holds for $\mathcal{E}^{\mathrm{raw}}_{h,D}(X)$.

Proof. Expand the square in (8) and interchange the finite sums. For $k = \ell$, the target argument is $r = abk^2$ and lemma 4.2 gives

$$HI_0(G)\log H \sum_{(a,b)=1} \sum_k u(ak)^2|W(abk^2/X)|^2 + O_G(H\mathcal{S}_D(X)).$$

The double source sum is exactly $\mathcal{S}_D(X)$: the substitution $n = ak, m = bk$ is unique. Inserting lemma 4.3 produces the main term in (27). All diagonal errors are $O_{\varepsilon,C,W,G}(HX(\log X)^2)$.

For $k \neq \ell$, the two target arguments

$$r = abk^2, \qquad r' = ab\ell^2$$

are distinct. Apply lemma 4.4, take absolute values of the source coefficients, and then use lemma 4.5. The total off-diagonal contribution is

$$\ll_{C,W,G} H\log\log(3X)\,\mathcal{L}_D(X) \ll_{C,W,G} HX(\log X)^2\log\log(3X).$$

This proves the centered theorem. For the raw energy, the target diagonal Λ^2 has the same leading term by lemma 4.2, and the same upper-sieve ledger handles the off-diagonal. □

Remark 4.7 (Where the saving comes from)*.* No cancellation is asserted in an individual same-ray prime correlation. The target diagonal contributes $H\log H$, while the absolute off-diagonal upper sieve contributes only $H\log\log X$. The source diagonal is of order $X\log^2 X$, and the entire source off-diagonal ledger has the same order. The extra logarithm on the target diagonal is the sole reason an asymptotic results.

5 Exact Fejér resolution and hard-band asymptotics

The sharp interval in theorem 3.1 leaves an $O(X\mathcal{E})$ boundary error. A kernel whose Fourier transform has compact support removes that error exactly.

Define the normalized Fejér kernel

$$\kappa(t) := \frac{2}{\pi}\left(\frac{\sin(t/2)}{t}\right)^2, \qquad \kappa(0) := \frac{1}{2\pi}. \tag{28}$$

With the convention $\widehat{\kappa}(\xi) = \int_{\mathbb{R}} \kappa(t)e^{-it\xi}dt$, one has

$$\widehat{\kappa}(\xi) = (1-|\xi|)_+, \qquad \int_{\mathbb{R}} \kappa(t)dt = 1. \tag{29}$$

Theorem 5.1 (Exact factor-ray Parseval identity)**.** *If $T\delta_X \geq 1$, where δ_X is defined in (9), then*

$$\int_{\mathbb{R}} \kappa(t/T)|\mathcal{R}_{h,D}(X;t)|^2 dt = T\mathcal{E}_{h,D}(X). \tag{30}$$

The same identity holds for the raw transform and raw ray energy.

Proof. Insert (6), expand the square, and use (29):

$$\int_{\mathbb{R}} \kappa(t/T)e^{it(\lambda_\rho - \lambda_{\rho'})}dt = T\widehat{\kappa}(T(\lambda_{\rho'} - \lambda_\rho)).$$

For distinct rays the argument has absolute value at least $T\delta_X \geq 1$, so the triangle function vanishes, including at the endpoints. Every diagonal term has multiplier T. □

Thus an effective bandwidth $T \asymp_W X$ exactly resolves all primitive factor rays. It does not resolve different k inside one ray; those terms have already been combined into $c_{a,b}$.

Corollary 5.2 (Averaged-shift Fejér-band asymptotic)**.** *Under the hypotheses of theorem 4.6, suppose also that $T\delta_X \geq 1$. Then*

$$\begin{aligned} &\sum_h G(h/H) \int_{\mathbb{R}} \kappa(t/T)|\mathcal{T}_{h,D}(X;t) - \mathcal{M}_D(X;t)|^2 dt \\ &\quad = \frac{1}{2} THXI_0(G)I_2(W)\log H\big((\log X)^2 - (\log D)^2\big) \\ &\qquad + O_{\varepsilon,C,W,G}\Big(THX(\log(2X))^2 \log\log(3X)\Big). \end{aligned} \tag{31}$$

If $G \not\equiv 0$ and $W \not\equiv 0$, this is an asymptotic throughout the stated D, H range.

Proof. Sum (30) with weight $G(h/H)$ and insert (27). Since $D \leq X^{1-\varepsilon}$ and $H \geq X$, the main term is $\gg_{\varepsilon,G,W} THX(\log X)^3$ when $I_0(G)I_2(W) > 0$, whereas the error is smaller by a factor $O(\log\log X/\log X)$. □

Corollary 5.3 (Averaged-shift hard-band formula)**.** *Under the hypotheses of theorem 4.6, for every $T \geq 1$,*

$$\begin{aligned} &\sum_h G(h/H) \int_{-T}^{T} |\mathcal{T}_{h,D}(X;t) - \mathcal{M}_D(X;t)|^2 dt \\ &\quad = THXI_0(G)I_2(W)\log H\big((\log X)^2 - (\log D)^2\big) \\ &\qquad + O_{\varepsilon,C,W,G}\Big(THX(\log(2X))^2 \log\log(3X) + HX^2(\log(2X))^3\Big). \end{aligned} \tag{32}$$

Consequently (32) is an asymptotic whenever $T/X \to \infty$.

Proof. Sum (11) over h. Twice the leading term in (27) gives the displayed main term. The first error is twice T times the averaged ray-energy error. For the Montgomery–Vaughan boundary error, theorem 4.6 and its proof give

$$\sum_h G(h/H)\mathcal{E}_{h,D}(X) \ll_{\varepsilon,C,W,G} HX(\log X)^3.$$

Multiplication by X gives the second error in (32). Relative to the main term, the two errors are respectively $O(\log\log X/\log X)$ and $O(X/T)$. $\square$

Corollary 5.4 (The TPC–10 truncation)**.** *Let*

$$D = X^{1/2}(\log X)^{-B}$$

with fixed $B > 0$*. Then*

$$(\log X)^2 - (\log D)^2 = \left(\frac{3}{4} + o(1)\right)(\log X)^2. \tag{33}$$

Thus the leading coefficient contributed by the source cutoff is $3/8$ *in* (31) *and* $3/4$ *in* (32)*, before the common factors* $THXI_0(G)I_2(W)\log H(\log X)^2$ *are restored.*

Remark 5.5 (Raw target). By the raw part of theorem 4.6, both band corollaries remain valid with $\mathcal{T}_{h,D}$ in place of $\mathcal{T}_{h,D} - \mathcal{M}_D$. Centering is used because $\mathcal{M}_D$ is the long-shift first-moment model, not because it changes the leading second-moment scale.

5.1 A parity-conditioned even-shift ensemble

The all-shift average above is not forced by the proof. To isolate the local parity factor without selecting an individual shift, define

$$\mathcal{M}_D^{(2)}(X;t) := 2 \sum_{\substack{n>D,\ m\geq 1 \\ 2\nmid mn}} u(n)W(mn/X)(n/m)^{it} \tag{34}$$

For even h, put $\mathcal{R}_{h,D}^{(2)} = \mathcal{T}_{h,D} - \mathcal{M}_D^{(2)}$ and define

$$\eta_r^{(2)}(h) := \Lambda(r+h) - 2\mathbf{1}_{2\nmid r}. \tag{35}$$

Then the ray coefficient of $\mathcal{R}_{h,D}^{(2)}$ is

$$c_{a,b}^{(2)}(h;X,D) = \sum_{\substack{k\geq 1 \\ ak>D}} u(ak)\eta_{abk^2}^{(2)}(h)W(abk^2/X), \tag{36}$$

and $\mathcal{E}_{h,D}^{(2)}$ denotes the sum of its squared moduli. The factor 2 in (34) is fixed by the first-moment identity

$$\frac{2}{H}\sum_{h\equiv 0\ (2)} G(h/H)\mathcal{T}_{h,D}(X;t) = I_0(G)\mathcal{M}_D^{(2)}(X;t) + O_{A,C,W,G}\Big(X(\log X)^{-A}\Big), \tag{37}$$

uniformly for real t and $X \leq H \leq X^C$.

Proposition 5.6 (Even-shift ensemble)**.** *Under the hypotheses of theorem 4.6, the ray energy* $\mathcal{E}^{(2)}_{h,D}$ *of* $\mathcal{R}^{(2)}_{h,D}$ *satisfies*

$$\begin{aligned}\sum_{h\equiv 0\ (2)} G(h/H)\mathcal{E}^{(2)}_{h,D}(X) = {} & \frac{1}{4}HXI_0(G)I_2(W)\log H \\ & \times \left((\log X)^2 - (\log D)^2\right) \\ & + O_{\varepsilon,C,W,G}\Big(HX(\log X)^2\log\log(3X)\Big).\end{aligned} \tag{38}$$

Consequently the hard-band leading coefficient is one half of that in (32)*, and the Fejér-band leading coefficient is one half of that in* (31)*.*

Proof. For $r \in [\alpha X, \beta X]$, the prime number theorem modulo 2 gives

$$\sum_{h\equiv 0\ (2)} G(h/H)\Lambda(r+h) = HI_0(G)\mathbf{1}_{2\nmid r} + O_A(H(\log H)^{-A}). \tag{39}$$

Summing this estimate against the source rows proves (37). The same argument for Λ^2, followed by expansion of (35), gives

$$\sum_{h\equiv 0\ (2)} G(h/H)|\eta^{(2)}_r(h)|^2 = HI_0(G)\log H\,\mathbf{1}_{2\nmid r} + O_{C,W,G}(H). \tag{40}$$

Indeed, for odd r the conditional density of primes in the odd residue class compensates for sampling half of the shifts. For even r, the target is even and $\Lambda(r+h)$ is supported only on powers of two, so its contribution is absorbed by the error.

For distinct $r, r' \in [\alpha X, \beta X]$, one also has

$$\sum_{h\equiv 0\ (2)} G(h/H)|\eta^{(2)}_r(h)\eta^{(2)}_{r'}(h)| \ll_{C,W,G} H\log\log(3X). \tag{41}$$

If r and r' are odd, write $h = 2j$ and apply the two-linear-forms Selberg upper sieve to $2j + r$ and $2j + r'$. If either is even, its target factor is supported on powers of two; the resulting polylogarithmic contribution is smaller than the displayed bound. This also covers the mixed-parity rows, which are present in (36).

The odd-product source diagonal is

$$\sum_{\substack{n>D,\ m\geq 1 \\ 2\nmid mn}} u(n)^2|W(mn/X)|^2 = \frac{1}{4}XI_2(W)((\log X)^2 - (\log D)^2) + O_W(X\log X). \tag{42}$$

Indeed, the odd m sum contributes one half of the continuous density, whereas removing the prime 2 does not change the leading $\frac{1}{2}\log^2 y$ term in the weighted $u(n)^2$ sum over odd n.

Expand the squared ray coefficients. Equations (40) and (42) give the displayed main term; even target diagonal rows contribute only $O(H\mathcal{S}_D(X))$. For $k \neq \ell$, apply (41) and then the full source ledger (26). The resulting error is $O(HX(\log X)^2\log\log(3X))$, proving (38). The band statements follow from the same factor-ray identities. □

Remark 5.7 (No individual-shift selection). The ensemble in proposition 5.6 contains only admissible parity shifts, but its exceptional behavior may still be concentrated at any prescribed even shift. No individual shift is selected.

6 Determinant layers and the mesoscopic boundary

The large-sieve theorem becomes diagonal only after an $O(X)$ resolution loss. This section records exactly what survives below that scale.

For a sharp band put

$$K_T(x) := \begin{cases} 2\sin(Tx)/x, & x \neq 0, \\ 2T, & x = 0. \end{cases} \tag{43}$$

Direct expansion of the original factor pairs gives

$$\begin{aligned} \int_{-T}^{T} |\mathcal{R}_{h,D}(X;t)|^2 dt = &\sum_{\substack{n,m,n',m' \\ n,n'>D}} u(n)u(n')\eta(mn+h)\eta(m'n'+h) \\ &\times W(mn/X)W(m'n'/X)K_T\left(\log \frac{nm'}{mn'}\right), \end{aligned} \tag{44}$$

where $\eta(q) = \Lambda(q) - 1$. The highest-degree von Mangoldt term in (44) contains

$$\Lambda(n)\Lambda(n')\Lambda(mn+h)\Lambda(m'n'+h).$$

Thus a small fixed-shift band is a genuine fourfold arithmetic correlation, not merely a divisor mean value.

After ray compression, the same exact identity is

$$\int_{-T}^{T} |\mathcal{R}_{h,D}(X;t)|^2 dt = \sum_{\substack{(a,b)=1 \\ (c,d)=1}} c_{a,b}\overline{c_{c,d}} K_T\left(\log \frac{ad}{bc}\right). \tag{45}$$

The relevant near-resonance is measured by the integer

$$\Delta((a,b),(c,d)) := ad - bc. \tag{46}$$

Proposition 6.1 (Compact-Fourier determinant layers)**.** *Let $\Phi \in L^1(\mathbb{R})$ and suppose, with the Fourier convention used in (29), that* $\operatorname{supp}\widehat{\Phi} \subset [-L, L]$*. Then*

$$\begin{aligned} \frac{1}{T}\int_{\mathbb{R}} \Phi(t/T)|\mathcal{R}_{h,D}(X;t)|^2 dt = &\sum_{\substack{(a,b)=1 \\ (c,d)=1}} c_{a,b}\overline{c_{c,d}} \\ &\times \widehat{\Phi}\left(T \log \frac{bc}{ad}\right). \end{aligned} \tag{47}$$

If $L/T \leq 1$, every nonzero summand satisfies

$$|ad - bc| \leq 2\beta X \sinh\left(\frac{L}{2T}\right) \ll_L \frac{X}{T}. \tag{48}$$

Proof. Insert (6) and use Fourier inversion for the scaled weight. A term can survive only if

$$\left|\log \frac{ad}{bc}\right| \leq \frac{L}{T}.$$

The exact hyperbolic identity from lemma 2.3 gives

$$|ad - bc| = 2\sqrt{abcd}\,\sinh\left(\frac{1}{2}\left|\log \frac{ad}{bc}\right|\right).$$

Since $ab, cd \leq \beta X$, this is at most the right-hand side of (48). □

At frequency scale $T = X^{\theta}$, the compact-Fourier mean square therefore contains only determinant layers

$$|\Delta| \ll X^{1-\theta}.$$

For $T \gg X$, only $\Delta = 0$ remains; primitive fractions then force $(a,b) = (c,d)$. At $T \asymp X$, only finitely many nonzero determinant layers can remain. For $T = o(X)$, the permitted determinant range grows and the number of potentially surviving layers is allowed to grow. A diagonal law then requires new cancellation among their arithmetic coefficients.

Proposition 6.2 (Sharpness of the X threshold)**.** *Let $\Phi \in L^1(\mathbb{R})$ have continuous Fourier transform with $\widehat{\Phi}(0) \neq 0$. There are hyperbolic coefficient systems supported on two primitive rays for which the Φ-weighted form has a nonzero off-diagonal term whenever $1 \leq T \leq cX$, with $c > 0$ depending only on the kernel and support band. Consequently the order $T \asymp X$ in theorem 5.1 cannot be uniformly improved for arbitrary coefficients.*

Proof. Take the two rays $(N,1)$ and $(N-1,1)$ with $N \asymp X$, and assign both coefficient one. Their determinant is 1 and their frequency gap is $\log(N/(N-1)) = N^{-1} + O(N^{-2})$. Continuity and $\widehat{\Phi}(0) \neq 0$ give an interval about zero on which $\widehat{\Phi}$ is nonzero. Choose c so that every scaled gap with $1 \leq T \leq cX$ lies in that interval. The cross term in (47) then survives. $\square$

6.1 The fixed-shift arithmetic left inside one ray

The diagonal ray energy is not an elementary coefficient norm. Expanding one term gives

$$\begin{aligned} |c_{a,b}|^2 = \sum_{k,\ell} & u(ak)u(a\ell)\eta(abk^2+h)\eta(ab\ell^2+h) \\ & \times W(abk^2/X)W(ab\ell^2/X), \end{aligned} \tag{49}$$

with the source cutoff understood. Terms $k \neq \ell$ contain two target prime weights on the quadratic sequences

$$abk^2 + h, \qquad ab\ell^2 + h.$$

For example, the ray $(a,b) = (1,1)$ has coefficient

$$c_{1,1}(h;X,D) = \sum_{k>D} u(k)\eta(k^2+h)W(k^2/X). \tag{50}$$

No fixed-h asymptotic for (50) or for the general off-diagonal (49) is used or obtained.

The role of the long shift average is now transparent. It does not make the same-ray correlations disappear. Rather, their total absolute contribution is upper-bounded, while the $k = \ell$ target square acquires the larger factor $\log H$. Specializing the averaged theorem back to one fixed shift would remove precisely that advantage.

Remark 6.3 (The next analytic target). A genuine advance into the mesoscopic region $1 \ll T \ll X$ would estimate the finitely many determinant-layer forms in (47) using their arithmetic coefficients, rather than their absolute values. Possible honest weakenings include averaging h over shorter windows, averaging the determinant Δ, or restricting (a,b) to balanced subregions. The present paper supplies the exact layer identity, the permitted determinant range, and the normalization against which such an estimate should be measured.

7 Endpoint recovery and the fixed-shift strength threshold

The ratio transform does not select $m = 1$ by merely taking a long sharp frequency interval. A fixed scale window does select it, but that window uses a coherent superposition of all frequencies.

Assume in this section that

$$\operatorname{supp} W \subset [\alpha, \beta], \qquad \beta < 4\alpha, \qquad D < \alpha X. \tag{51}$$

Choose $V \in C_c^\infty(\mathbb{R})$ such that

$$V(x) = 1 \quad (\log \alpha \le x \le \log \beta), \qquad V(x) = 0 \quad (x \le \log(\beta/4)). \tag{52}$$

Proposition 7.1 (Exact one-Mellin endpoint selector)**.** *Under* (51)*,*

$$\sum_n u(n)\Lambda(n+h)W(n/X) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{V}(t)X^{-it}\mathcal{T}_{h,D}(X;t)dt, \tag{53}$$

$$\sum_n u(n)(\Lambda(n+h)-1)W(n/X) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{V}(t)X^{-it}\mathcal{R}_{h,D}(X;t)dt, \tag{54}$$

where $\widehat{V}(t) = \int_{\mathbb{R}} V(x)e^{-itx}dx$*.*

Proof. If $r = mn$ lies in the support of $W(r/X)$, then

$$\log(n/m) - \log X = \log(r/X) - 2\log m. \tag{55}$$

For $m = 1$ this lies in $[\log \alpha, \log \beta]$, where $V = 1$. For $m \ge 2$ it is at most $\log \beta - 2\log 2 = \log(\beta/4)$, where $V = 0$. Thus inserting $V(\log(n/m) - \log X)$ selects precisely $m = 1$. The cutoff $D < \alpha X$ removes none of those rows. Fourier inversion of V gives (53); subtracting the same identity for (4) gives (54). □

Remark 7.2 (General product supports). Every compact subinterval of $(0, \infty)$ is a finite union of intervals of multiplicative width less than four. A smooth partition of unity in the product variable therefore reduces a general W to finitely many selectors of the form above.

7.1 Truncating the coherent selector

The total variation of either factor-scale measure in (53)–(54) satisfies

$$\sum_{\substack{n>D \\ m\ge 1}} |u(n)|(1 + \Lambda(mn+h))|W(mn/X)| \ll_{h,W} X\log(2X). \tag{56}$$

One way to see this is to put $r = mn$, use

$$\sum_{n|r} |u(n)| \le \log r + \tau(r),$$

and apply the prime number theorem to the logarithmic part and the standard shifted Titchmarsh upper bound to the divisor part [1].

Since $V \in C_c^\infty$, for every $A > 0$,

$$\left|\int_{|t|>T} \widehat{V}(t)X^{-it}\mathcal{R}_{h,D}(X;t)dt\right| \ll_{A,h,V,W} X\log(2X)T^{-A}. \tag{57}$$

The same estimate holds for the raw transform and for the source model. Thus any positive power of $\log X$ is a sufficient truncation length for an $o(X)$ approximation, after choosing A large. The hard part is not the smooth Fourier tail; it is the accuracy of the arithmetic model on the retained band.

7.2 An exact L^2 strength test

For a function F for which the integral converges, define

$$\mathscr{L}_X(F) := \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{V}(t)X^{-it}F(t)dt. \tag{58}$$

Proposition 7.3 (Endpoint discrepancy forces band energy)**.** *Let S and M be two transforms and let*

$$\mathrm{Tail}_T(S-M) := \frac{1}{2\pi}\left|\int_{|t|>T} \widehat{V}(t)X^{-it}(S(t)-M(t))dt\right|. \tag{59}$$

Then

$$\|S-M\|_{L^2[-T,T]} \geq \frac{2\pi}{\|\widehat{V}\|_{L^2(\mathbb{R})}}\left(|\mathscr{L}_X(S)-\mathscr{L}_X(M)| - \mathrm{Tail}_T(S-M)\right)_+. \tag{60}$$

Consequently, if the endpoint discrepancy is $\gg X$ and the Fourier tail is $o(X)$, then

$$\int_{-T}^{T} |S(t)-M(t)|^2 dt \gg X^2. \tag{61}$$

Conversely, an absolute band error $o(X^2)$ together with an $o(X)$ tail forces the two endpoint functionals to agree up to $o(X)$.

Proof. Split (58) at $|t|=T$, apply the triangle inequality, and then apply Cauchy–Schwarz on $[-T,T]$:

$$|\mathscr{L}_X(S)-\mathscr{L}_X(M)| \leq \frac{\|\widehat{V}\|_2}{2\pi}\|S-M\|_{L^2[-T,T]} + \mathrm{Tail}_T(S-M).$$

Rearrangement proves (60). The two consequences are immediate. □

The normalization is important. A statement

$$\int_{-T}^{T} |S-M|^2 dt = o(TX^2)$$

is only a root-mean-square statement. When $T\to\infty$, Cauchy–Schwarz allows an $o(X\sqrt{T})$ coherent error and does not recover the endpoint. The absolute $o(X^2)$ threshold in proposition 7.3 is strictly stronger.

7.3 The Hardy–Littlewood boundary

For a nonzero integer shift h, write the usual prime-pair singular series as

$$\mathfrak{S}(h) := \begin{cases} 2C_2 \prod\limits_{\substack{p\mid h\\ p>2}} \dfrac{p-1}{p-2}, & 2\mid h,\\ 0, & 2\nmid h, \end{cases} \qquad C_2 := \prod_{p>2}\left(1-\frac{1}{(p-1)^2}\right). \tag{62}$$

This is the constant in the corresponding Hardy–Littlewood prediction [3]; no such prediction is assumed below.

For the raw prime-target transform, the selected endpoint is

$$\begin{aligned} \mathscr{L}_X(\mathcal{T}_{h,D}) &= \sum_n (\Lambda(n)-1)\Lambda(n+h)W(n/X) \\ &= \sum_n \Lambda(n)\Lambda(n+h)W(n/X) - XI_0(W) + o_{h,W}(X), \end{aligned} \tag{63}$$

by the shifted prime number theorem. Hence, for one fixed admissible even shift and one fixed W, evaluating (63) with the predicted main term is equivalent to the corresponding single-weight Hardy–Littlewood assertion. The centered endpoint has the same predicted leading term because $\sum_n (\Lambda(n)-1)W(n/X) = o_W(X)$.

It follows from proposition 7.3 that a fixed-shift band model with absolute $o(X^2)$ error, a compatible filtered model value, and $o(X)$ tails would already contain that Hardy–Littlewood endpoint. The implication is one-way: the endpoint controls one filtered functional, not the entire band.

There is an unconditional parity check on the natural source model.

Corollary 7.4 (Odd-shift model floor)**.** *Assume $I_0(W) \neq 0$ and fix an odd integer h. Let $S = \mathcal{T}_{h,D}$ and $M = \mathcal{M}_D$. If $T \to \infty$ so that the tail in (59) is $o(X)$—for example any fixed positive power of $\log X$—then*

$$\int_{-T}^{T} |\mathcal{T}_{h,D}(X;t) - \mathcal{M}_D(X;t)|^2 dt \gg_{h,V,W} X^2. \tag{64}$$

Proof. If h is odd, two prime-power-supported values n and $n+h$ can both occur only when one is a power of two. There are $O(\log X)$ such possibilities, so

$$\sum_n \Lambda(n)\Lambda(n+h)W(n/X) = o_{h,W}(X).$$

The prime number theorem applied to the two linear terms in the centered endpoint then gives

$$\mathscr{L}_X(\mathcal{T}_{h,D} - \mathcal{M}_D) = -XI_0(W) + o_{h,W}(X).$$

Apply proposition 7.3 and (57). □

Remark 7.5 (Why this is not a twin-prime lower bound)*.* The odd-shift corollary is a local-parity sanity check: the target-free model does not encode the singular local factor. For admissible even shifts, replacing the explicit odd-shift value by $(\mathfrak{S}(h)-1)XI_0(W)$ is exactly the unproved prime-pair step. No such replacement is made here.

8 Finite certificate and reproducibility

The proofs above are analytic. A deterministic finite certificate is included only to test the exact algebra that precedes the estimates. It does not test a prime-pair asymptotic. The script `experiments/factor_ray_certificate.py` performs the following checks for a documented finite hyperbolic support.

(i) It enumerates every supported factor pair (n,m) and groups it by the reduced fraction $(a,b) = (n/(n,m), m/(n,m))$. At several real frequencies it compares the original factor-pair sum with the compressed ray sum.

(ii) It computes all adjacent gaps between the ordered ray frequencies (and hence their global minimum), verifies the exact hyperbolic lower bound in (9), and records the minimizing rays.

(iii) It evaluates the hard-band quadratic form in two ways: directly from the factor pairs and from the compressed ray coefficients. It also verifies that the Fejér multiplier is diagonal once the finite frequency bandwidth exceeds the reciprocal of the actual minimum ray gap.

(iv) It counts the determinant layers permitted by the compact-Fourier support condition at a sequence of resolutions and verifies the endpoint classification $m = 1$ versus $m \geq 2$ under (51).

The companion unit test repeats the certificate at smaller parameter values and uses exact rational arithmetic for the ray labels and determinants. Floating-point comparisons occur only in exponential and logarithmic evaluations and are guarded by reported tolerances. The machine-readable output is `experiments/data/factor-ray-certificate.json`. Reproduction requires only the Python standard library:

```
python experiments/factor_ray_certificate.py
python -m unittest experiments.test_factor_ray_certificate
```

The saved output is a regression certificate for the identities, not evidence for the fixed-h statements explicitly excluded in section 1.

9 Conclusion and next boundary

This paper closes one well-defined step in the factor-scale program. A complete large-factor prime-target residual is converted into a separated exponential polynomial by grouping factor pairs along primitive ratio rays. At fixed shift this yields a hyperbolic Mellin large-sieve law, an $X \log^4 X$ ray-energy upper bound, density-one cancellation in long frequency intervals, and exact Fejér diagonalization at the geometric resolution $T \asymp X$. None of these conclusions evaluates a prescribed frequency.

Squaring first and averaging h over a window of length at least X adds arithmetic content. The source diagonal admits an explicit asymptotic evaluation; the target square contributes $\log H$; and a two-linear-forms upper sieve is sufficient for every same-ray off-diagonal. The result is an unconditional averaged-shift ray-energy asymptotic and its hard- and smooth-band consequences. The even-shift version records the parity-conditioned local factor without selecting any individual even shift.

The remaining fixed-shift obstruction is now explicit rather than hidden in a generic appeal to the parity problem. Spectral separation removes different reduced ratios but leaves all radial indices k coherent. Their coefficients contain quadratic target correlations $\Lambda(abk^2+h)\Lambda(ab\ell^2+h)$. In the mesoscopic band, additional interactions are organized by the determinant $ad-bc$. At the factor endpoint, a single smooth coherent functional recovers the weighted Hardy–Littlewood quantity of Hardy and Littlewood [3]; hence an absolute $o(X^2)$ fixed-shift band approximation, together with a compatible filtered model and $o(X)$ tails, would already have substantial prime-pair content.

The next honest target is therefore not to assert a fixed-h diagonal formula. It is to find cancellation in one controlled weakening of the determinant-layer or intra-ray forms: a shorter shift average, a balanced ray sector, or an average over a bounded set of determinant layers. Any such theorem must be compared with the endpoint strength test before it is interpreted arithmetically. The classical sieve framework described by Iwaniec and Kowalski [4] remains the relevant boundary: no twin-prime lower bound, Hardy–Littlewood asymptotic at $h=2$, or violation of sieve parity is obtained here.

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